

$$\begin{array}{ccccccc}
 \mathbf{L}^2(\mathbb{R}^2) & \longrightarrow & \cdots & \begin{array}{c} \nearrow \\ \searrow \end{array} & V_{j-1} \otimes V_{j-1} & \begin{array}{c} \nearrow \\ \searrow \end{array} & V_j \otimes V_j & \begin{array}{c} \nearrow \\ \searrow \end{array} & V_{j+1} \otimes V_{j+1} & \begin{array}{c} \nearrow \\ \searrow \end{array} & \cdots & \longrightarrow & \{0\} \\
 & & & \searrow & & & \searrow & & \searrow & & \searrow & & & \\
 & & & & W_{j-1}^{(2)} & & W_j^{(2)} & & W_{j+1}^{(2)} & & & & & 
 \end{array}$$